Workshop Project Brief: Leave Request Module

This brief explains the project you will build in Session 2: the problem, who uses the system, how a request moves, what is in scope and which tools you will use.

1. At a glance

Item	Detail
Project	Leave Request module for an HR Management System (HRMS)
Company	Budiman Company, about 200 employees (a fictional company)
Users	Employee, Supervisor, HR and Admin
What you build	A web app where a leave request moves from the employee to the supervisor to HR, with its status visible at every step
How you build it	Spec-Driven Development: write and review the specifications first, then the AI builds from them
Where you start	A starter project with a working login for all four roles

2. The scenario

Budiman Company still processes leave on paper. The employee fills in a leave form and hands it to their supervisor. The supervisor signs it, then HR receives a copy and enters it into a spreadsheet by hand.

The HR Manager has asked your team to replace the paper form with a Leave Request module in the company's HR Management System.

3. Problem statement

The paper process causes three problems:

1. Forms get lost or delayed between the employee, the supervisor and HR.
2. Employees can't see where their request is: waiting for the supervisor, with HR, or decided.
3. HR has no single list of requests waiting for a decision.

Goal: one request, one system, visible from submission to decision. Every person sees exactly the requests they need to act on, and nothing gets lost.

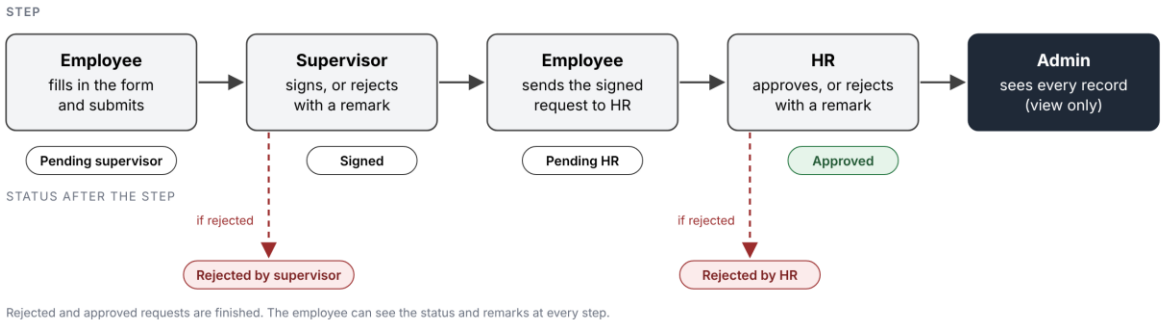
4. Who uses the system

Role	What they do

Employee	Submits a leave request to a supervisor in their own division, tracks its status and remarks, and sends a signed request to HR.
Supervisor	Sees the requests sent to them, and signs or rejects each one (a rejection needs a remark).
HR	Sees every request waiting for HR in one shared list, and approves or rejects it (a rejection needs a remark). The officer who decides is recorded.
Admin	Sees every leave record in the system. View only: changes nothing.

5. The workflow

A request moves through these steps. The status changes after each one.



Rules the workflow must follow:

1. Employees apply for leave; supervisors and HR only decide.
2. The employee chooses a supervisor from their own division.
3. Every request passes the supervisor first: signed, or rejected with a reason.
4. After it is signed, the employee sends it to the shared HR queue. Either HR officer may decide, and the one who acts is recorded.
5. The admin sees every request and changes nothing.
6. Every action (submit, sign, reject, send, approve) asks for confirmation first and shows a success message after.

6. Scope within this session

Role	What they do
Core	Employee submits → supervisor signs or rejects → employee sends to HR → HR approves or rejects → admin sees the record. Each role has a page with its current and past request.

7. Decisions you will make

The brief leaves these open on purpose. The AI will ask you about each one while writing the requirements, and different answers are welcome:

- Which leave types there are.
- Which fields the form has, including whether it asks for a reason at all.
- How days are counted: calendar days, working days or shifts.
- What each role's home page shows, and the columns of its tables.
- Where finished requests go.

8. What you are given

Item	What is it for
Starter project (on GitHub)	A working app with login for all four roles, each landing on its own page. You fork it and build the module on top. Don't rebuild the login.
Sample login spec (docs/specs/in the starter)	A finished specification for the login, as a worked example. Your specs follow its style.
Workshop project brief	The scenario and rules to give the AI when it writes the specs with you.

Demo accounts

The app shown in the Session 1 demo.

Role	Log in as
Employee	employee-1@budiman.com
Supervisor	supervisor-1@budiman.com
HR	hr-1@budiman.com
Admin	admin@budiman.com

9. Tool and technologies

Tool	What it does in the project
Illham Harness	The AI coding tool. It writes the specs with you, builds the code, runs commands and commits your work, asking permission first.
Ilham (model: ilham-flash)	It runs the AI model inside Brunei.
VS Code	Code editor, for looking at the project's files and the specs.
Node.js	Runs the web app on your laptop.

Next.js with TypeScript	The web framework: builds both the screens and the server logic in one project.
SQLite with **Prisma**	The database: a single file on your laptop. Prisma lets the code read and write it.
Tailwind CSS	Styles the screens.
Git and **GitHub**	Save points for your work, and the online copy of your fork. The AI commits and pushes for you.

10. How you will work: Spec-Driven Development

You agree what to build in writing first. The AI builds from the specs, and you check the result against them.

Spec file	Question it answers
01-requirements.md	**What** should the module do, and **why**? User stories and an acceptance checklist.
02-design.md	**How** is it built? Part A: the data (tables, fields, statuses, rules). Part B: the app.
03-tasks.md	In **what order** is it built, and how is each step checked?

The build loop, for each task

1. The AI builds the task from the specs.
2. You review it and check it in the browser, against the acceptance checklist.
3. The AI commits and pushes it to your GitHub fork.

11. Running the app

Open a terminal in the project folder:

Command	What it does
npm install	Install what the project needs (first time only).
cp .env.example .env	Create your settings file
npm run db:migrate	Create or updates the database tables.
npm run db:seed	Reset the demo users and sample data.
npm run dev	Start the app. Open http://localhost:3000 in your browser